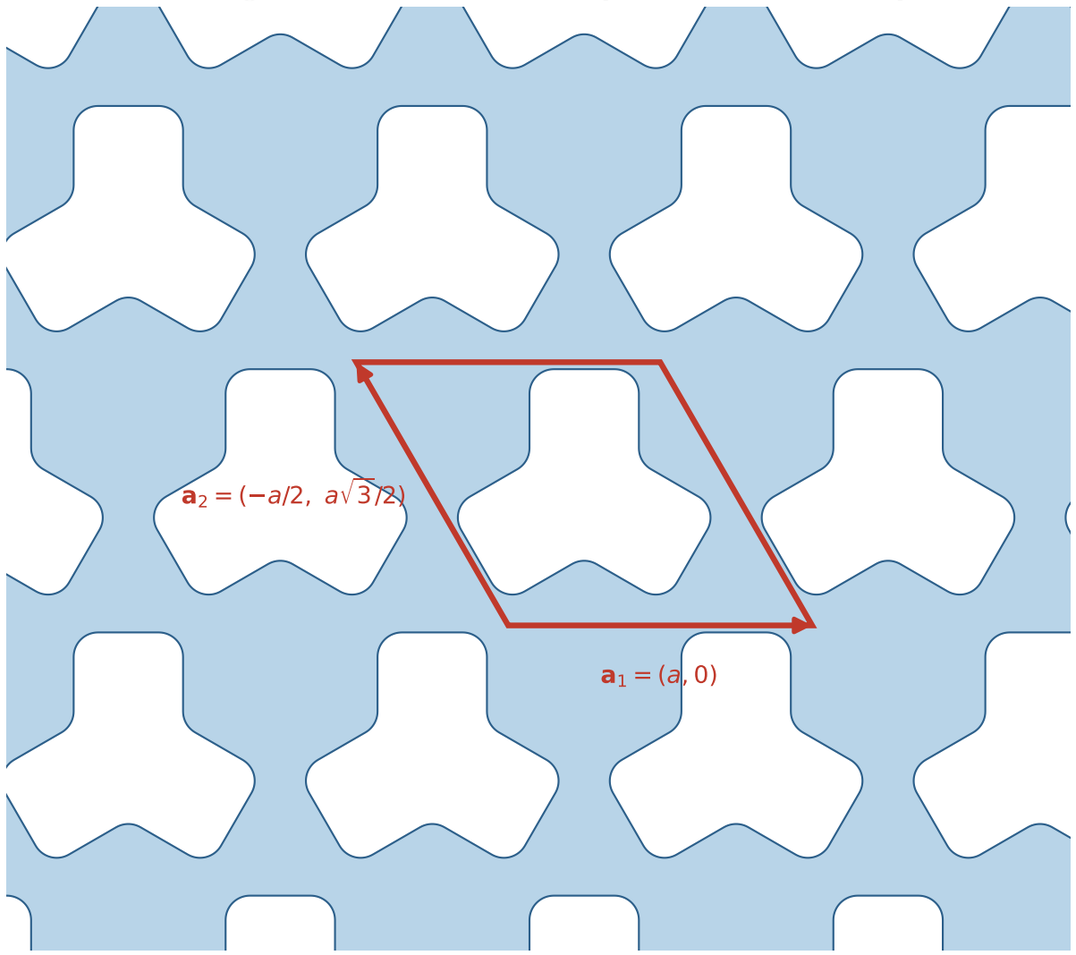
Diamond snowflake phononic crystal -- lattice-constant sweep (diamond_a_sweep)

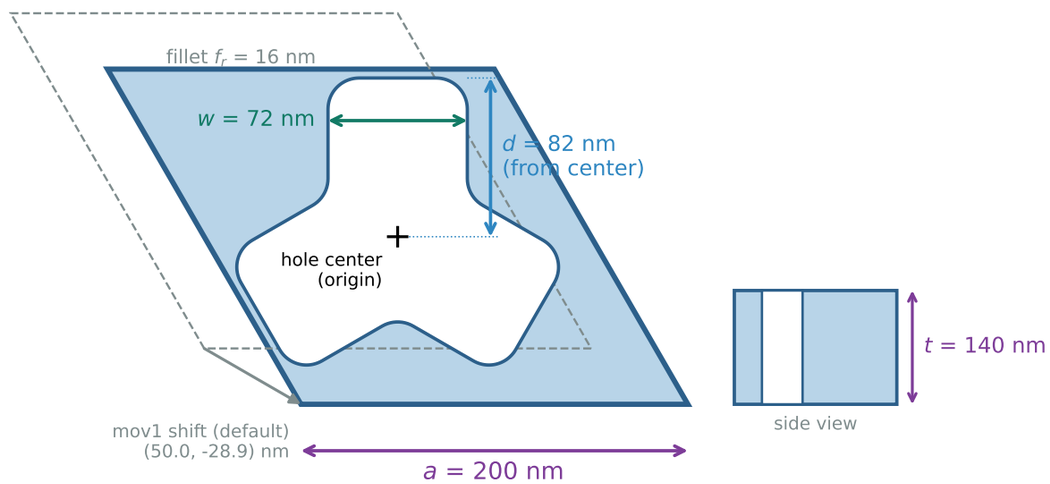
a = 150-275 nm (step 25) x d/a {0.35, 0.41, 0.48} x w/a {0.28, 0.36, 0.43} | t=140, f_r=16 nm fixed

E=1050 GPa, nu=0.07, rho=3500 kg/m^3 | N_mode=10, f_mech=10 GHz x 300/a | panels A-C drawn at a=200 nm, d/a=0.41, w/a=0.36

(A) Triangular lattice -- rhombus primitive cell (as in pol1)

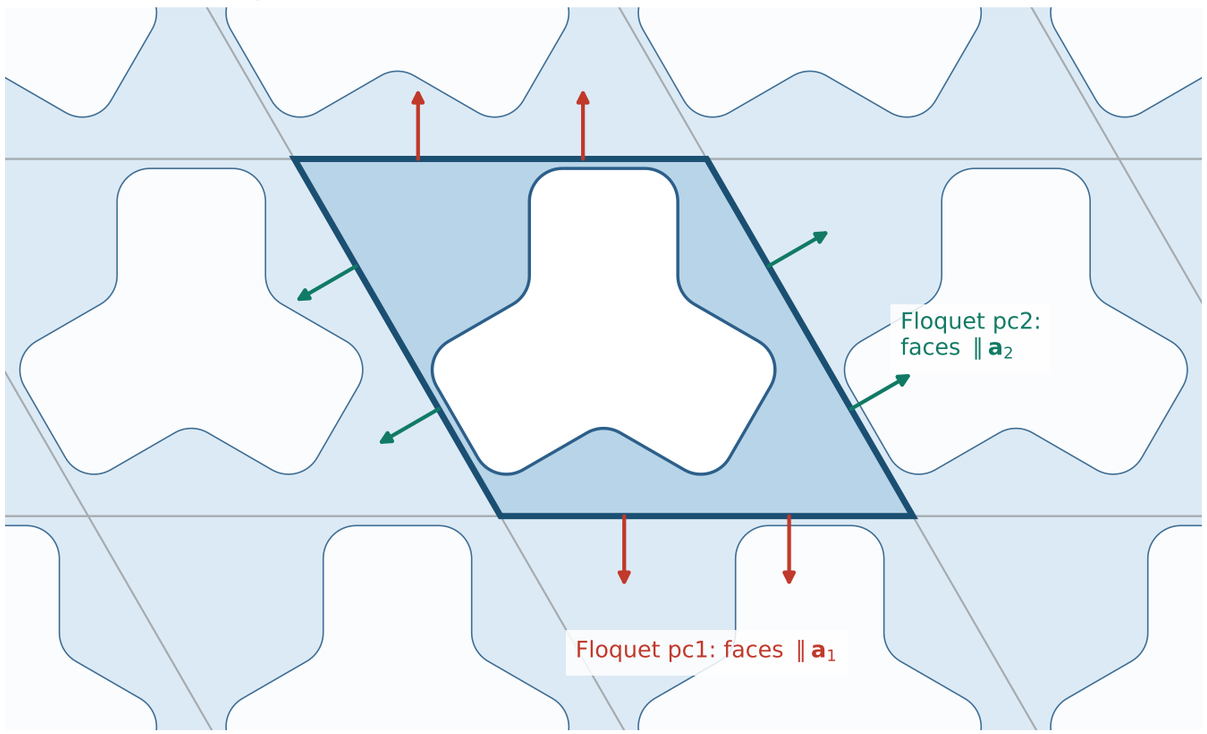


(B) Unit cell (top view): hole at lower-triangle centroid, 3 arms of length d from the center at 90/210/330 deg

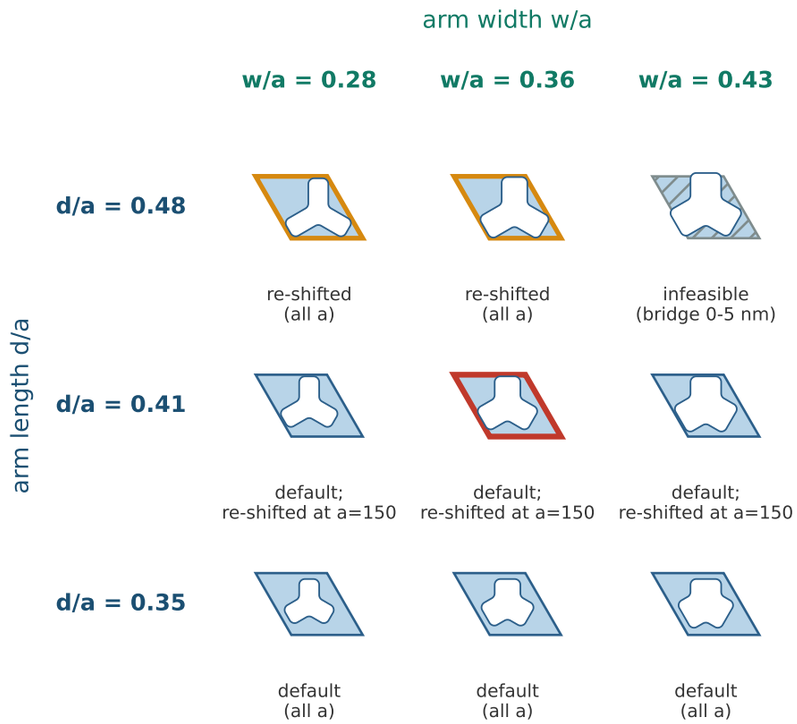


(C) COMSOL cell + Floquet BCs (pc1, pc2)

$\mathbf{k} = (k_{Fx}, k_{Fy}, 0)$ **Cartesian; z=0 midplane: sym1 / as1; top face free**



(D) Ratio grid at a=200 nm -- cell placement used



orange = cell re-shifted so the hole fits (physics-invariant, validated); hatched = cannot be modelled with this rhombus cell (hole meets its periodic images); red = panels A-C